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AE-LFOG-YOLO: robust safety helmet detection via adaptive anchors and illumination invariant learning

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In high-risk industrial environments such as tunnel construction, reliable safety helmet detection is critical for preventing head injuries. However, severe illumination inhomogeneity and multi-scale object appearances pose significant challenges to existing detectors due to static anchor designs and the absence of illumination-aware feature learning. This paper proposes AE-LFOG-YOLO, an end-to-end framework that enhances YOLOv8 through dual physics-informed optimizations. The approach integrates an Illumination-Invariant Module (IIM) that employs a dual-path feature decoupling strategy to suppress lighting artifacts within the network backbone. Concurrently, the Adaptive Evolutionary - Light Field Optimized Generation (AE-LFOG) algorithm replaces static anchors with a dynamic evolutionary process guided by local illumination gradients and thin-lens imaging principles, enabling continuous optimization of anchor parameters during training. Evaluated on a real-world tunnel dataset, the method achieves 94.83% mAP@0.5 and significantly improves robustness under challenging illumination variations, as evidenced by a 35.7% extension in effective operating range. These results demonstrate the effectiveness of integrating physical imaging priors into deep learning for robust visual perception in complex industrial scenarios.

Keywords YOLOv8, Safety helmet detection, Illumination robustness, Adaptive anchors, Physics-informed learning

The construction industry remains highly hazardous, with head injuries often stemming from falling objects and inadequate PPE use[1]. Safety helmets are critical for protection, yet compliance is frequently poor, motivating automated monitoring systems[2]. In challenging environments such as underground tunnels as Fig. 1, severe illumination variations including overexposure, shadows, and contrast gradients—degrade image quality and impair deep network feature learning, increasing false negatives[3].

To mitigate low-light degradation, physics-inspired enhancement methods have been proposed. Wang et al. developed a multi-scale Retinex unfolding network that integrates classical Retinex theory into a deep architecture for improved visual quality[4]. Du et al. introduced a hybrid zero-reference dehazing network, demonstrating its efficacy in enhancing images for safety helmet detection in mines[5]. Subsequent works like SRLFormer[6], CGRE-Retinex[7], and RetinexD-Net[8] combine Retinex with attention or dense connectivity for improved low-light enhancement, while YOLO[9] enables real-time detection through end-to-end processing and benefits from pre-training via transfer learning[10]. However, these approaches typically operate as pre-processing modules, creating a disjointed pipeline that decouples enhancement from detection and may hinder end-to-end optimization. The latest iteration, YOLOv8, has become a popular choice for industrial inspection tasks. Zhang et al. adapted YOLOv8 for helmet detection by refining network structure and post-processing strategies, achieving state-of-the-art performance[11]. Other variants focus on specific challenges: Liu et al. proposed YOLOv8-ME with an asymptotic feature pyramid for small-object detection in mining[12]. Chen et al. enhanced YOLOv8n with coordinate attention and lightweight heads for complex scenes[13]. Xu et al. introduced YOLO-GL with global-local attention for multi-scale detection[14] and Li et al. optimized YOLOv4-Tiny for edge deployment in remote sensing[15]. Despite these advances, critical limitations persist. Anchor box design remains largely static, relying on K-means clustering of training set dimensions[16], which fails to adapt to dynamic changes in object scale induced by perspective and illumination. Attention mechanisms such as an optimized YOLO NAS[17] significantly boosting real-time detection accuracy and robustness against occlusion and illumination

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Fig. 1. Typical Challenging Detection Scenarios in Engineering.

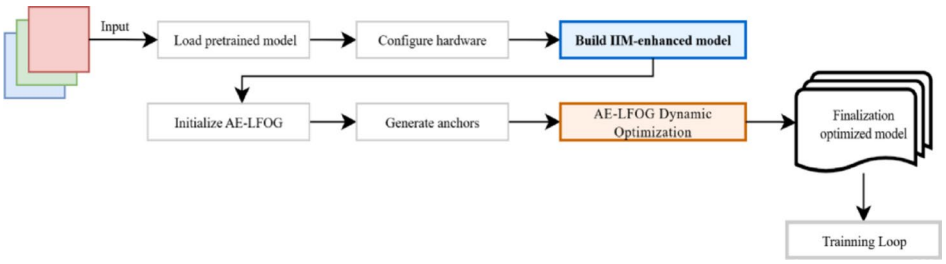


Fig. 2. Method Framework Diagram.

variations. While models like Shadow Hunter[18] address low-contrast conditions, they lack generalization across diverse environmental dynamics. Moreover, existing methods seldom jointly optimize illumination robustness and scale adaptability. For instance, although additional prediction heads can capture finer features for small objects[19], focus on enhancing feature representation for small targets, and a hybrid YOLOv8-ViT framework that leverages self-attention for global context modeling in complex scenes[20]. Chen et al. proposed WHU-YOLO, an end-to-end model integrating Ghost modules and Bi-FPN for efficient welding helmet detection, avoiding two-stage pipeline limitations[21]. Existing method in low-light conditions is limited by suboptimal architecture and feature integration, necessitating an end-to-end joint optimization approach.

To enhance reliability under complex illumination, we propose an improved YOLOv8 with two key components: an Illumination-Invariant Module (IIM) for adaptive feature recalibration without preprocessing, and Adaptive Evolutionary - Light Field Optimized Generation (AE-LFOG) for optical-flow-guided anchor optimization. Evaluated on the validation set, it achieves 94.83% mAP@0.5, reducing false alarms and missed detections, thereby enhancing operational safety in industrial settings.

Related work

Recent helmet detection research has focused on enhancing YOLO-based models through targeted architectural modifications. Baoju L et al.[22] replace standard convolutions with DSConv for efficiency and integrate Dysample for upsampling, adopting WIoU loss to accelerate convergence. Du Q. et al.[23] introduce the C2f-RFA module to refine features and propose CSP-EDPAN for multi-scale fusion. Han D. et al.[24] enhance YOLOv8 with SPD-Conv for detail preservation, a SEResNeXt head for feature refinement, and C2f-CA for small-target focus. Wang S. et al.[25] optimize the backbone by reducing channels, integrate Biformer for global-local context, add an LEF for low-light scenes, and use Power-IoU for faster training. Hu L. et al.[26] achieve lightweight performance via GhostNetv2 for efficient extraction and CA attention for small-target enhancement, combined with Focal-CIOU loss. Finally, a comparative study[27] benchmarks Swin Transformer as the top-performing backbone for YOLOv10 in complex real-world scenarios.The results of comparison with these approaches are presented as Table 4 in the experiments section.

Methodology

To address illumination inhomogeneity and multi-scale detection in complex scenes, this study propose AE-LFOG-YOLO integrating two physics-informed modules, as shown in Fig. 2, including IIM for lighting-robust feature extraction and AE-LFOG for dynamic anchor refinement. AE-LFOG updates anchors per epoch using data statistics and imaging priors, while IIM enhances backbone features. The design enables continuous adaptation to scale and illumination variations. Core logic is detailed in Algorithm 1.

```
Require: Dataset D, Epochs E, Initial anchors A_0, Base weights w_base, Scaling factors α_box, α_cls
Ensure: Trained model M with optimized anchors A_E and adaptive loss weights
A ← A_0; w_adjusted ← w_base  // Initialize
for epoch = 1 to E do
    for batch in DataLoader(D) do
        predictions ← M(images)  // Includes IIM forward pass
        loss ← ComputeLoss(predictions, targets, w_adjusted)
        loss.backward(); optimizer.step(); optimizer.zero_grad()
    end for
    A_candidate ← PerturbAnchors(A)
    if ScoreAnchorSet(A_candidate) > ScoreAnchorSet(A) then
        A ← A_candidate                    // From Eq.3
    end if
    S_optical ← GetOpticalScore(A)         // From Eq.8
    w_adjusted ← {L_box: w_base.L_box * (1 + α_box * (1 - S_optical)),
                  L_cls: w_base.L_cls * (1 + α_cls * S_optical),
                  L_df: w_base.L_df}       // Dynamic adjustment (Eq.10, 11)
end for
return M, A, w_adjusted
```

Algorithm 1. The AE-LFOG-YOLO Training Framework.

IIM enhanced module

Severe illumination variations in complex engineering construction scenarios can induce feature distribution shifts in standard convolutional neural networks. To address this, we propose the Illumination-Invariant Module (IIM), which enhances feature robustness without relying on external preprocessing. As shown in Fig. 3, the proposed IIM employs a dual-path feature decoupling mechanism to disentangle input features into $\mathrm{X}_{\mathrm{light}}$ and $X_{reflect}$. Components $\mathrm{X}_{\mathrm{light}}$ is a representation encoding illumination-sensitive patterns. Components $X_{reflect}$ is a structurally stable representation approximating reflectance-invariant properties. This decomposition enables the network to distinguish between appearance changes caused by lighting fluctuations and intrinsic object characteristics such as shape and texture.

A cascaded gating mechanism is employed to enhance illumination invariance. Given input features $\mathrm{X} \in \mathrm{R}^{\mathrm{C}\times\mathrm{H}\times\mathrm{W}}$, the global average pooled representation $\mathrm{g}=\mathrm{GAP}(\mathrm{X}) \in \mathrm{R}^{\mathrm{C}}$ is computed to capture channel-wise statistical characteristics.Based on *g*, a two-layer MLP with bottleneck structure generates the sensitivity scores $\mathrm{w}_{\mathrm{sens}}$, which reflect the degree to which each channel responds to illumination variations. In parallel, an identical network branch processes the same descriptor g to generate stability scores $\mathrm{w}_{\mathrm{stable}}$, capturing the reliability of each channel under varying lighting conditions. The scores are subsequently normalized via σ(·) that represents the Sigmoid function to constrain outputs to [0,1]. The output is as follows:

$$\mathrm{X}_{\mathrm{light}} = \mathrm{X} \odot [1 - \sigma(\mathrm{w}_{\mathrm{sens}})] \odot \sigma(\mathrm{w}_{\mathrm{stable}}) \tag{1}$$

A spatial structure extractor is employed to capture illumination-invariant geometric patterns. This sub-network first employs a 1 × 1 convolutional layer to project the input feature map *X* into a lower-dimensional space (C/2 channels).This is followed by a 3 × 3 convolution with batch normalization and ReLU activation to extract local structural cues such as edges and corners. Finally, another 1 × 1 convolution restores the channel dimension to C to produce the output $X_{reflect.}$ The spatial branch captures lighting-invariant structures to guide attention during fusion.

The two feature paths are integrated via channel-wise multiplication (⊙) to adaptively modulate the fused representation, as defined in Eq. (2). To enhance generalization and robustness, we introduce a residual connection combined with stochastic depth regularization: during training, the non-identity branch is randomly

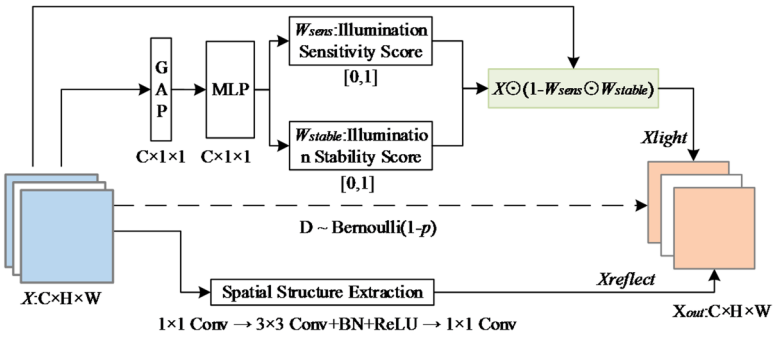


Fig. 3. Structure of the Illumination-Invariant Module (IIM).

skipped with probability p, while the identity mapping is preserved. Specifically, let $D \sim \text{Bernoulli}(1-p)$ is a binary mask. The output is computed as:

$$X_{out} = (1 - D) \cdot X + D \cdot (X_{reflect} \odot X_{light}) \quad (2)$$

As illustrated in Fig. 4, the overall network architecture maintains the structural organization of YOLOv8m while incorporating the proposed IIM enhancement modules. IIMs are inserted at key downsampling transitions to enhance photometric invariance under varying illumination. The neck and detection head employ standard YOLOv8 components (PAN-FPN and Detect), forming an end-to-end object detection pipeline.

AE-LFOG anchor optimization

In anchor-based detectors, anchor prior design critically affects performance. Conventional methods like K-Means[17] rely solely on GT box statistics, ignoring physical imaging principles. To align anchors with real-world optics, this paper proposes AE-LFOG which serves as a dynamic co-optimizer that refines anchors during training, bridging data-driven and physics-informed design.

AE-LFOG integrates optical constraints to shift anchor optimization toward physics-informed learning. At each epoch's end (Fig. 5), it collects ground truth boxes, evaluates candidate anchors via a composite score balancing data alignment and physical plausibility, and applies evolutionary selection to update anchors. The refined set feeds back into training, enabling priors to evolve into statistically representative and physically meaningful configurations. Details are provided in sections AE-LFOG framework and comprehensive scoring function respectively.

AE-LFOG framework design

The design of the AE-LFOG framework is inspired by biological evolution theory, formulating anchor optimization as an iterative process of perturbation, evaluation, and selection. This process is executed at the end of each training epoch to enable the anchor priors to dynamically adapt to the observed data distribution.

Perturbation introduces diversity into the current optimal anchor set $A_{current,}$ by applying random deviations to generate a new candidate set $A_{candidate}$. This enables the algorithm to escape local optima and adapt to extreme scale variations. The scoring function $S(A)$ which is detailed in next section evaluates both sets, quantifying how well anchors align with the data distribution and physical plausibility. If $S(A_{candidate}) > S(A_{current,})$, the candidate set replaces the incumbent. Through this iterative cycle, the anchor set dynamically evolves, ensuring effective coverage of diverse object scales in complex scenarios.

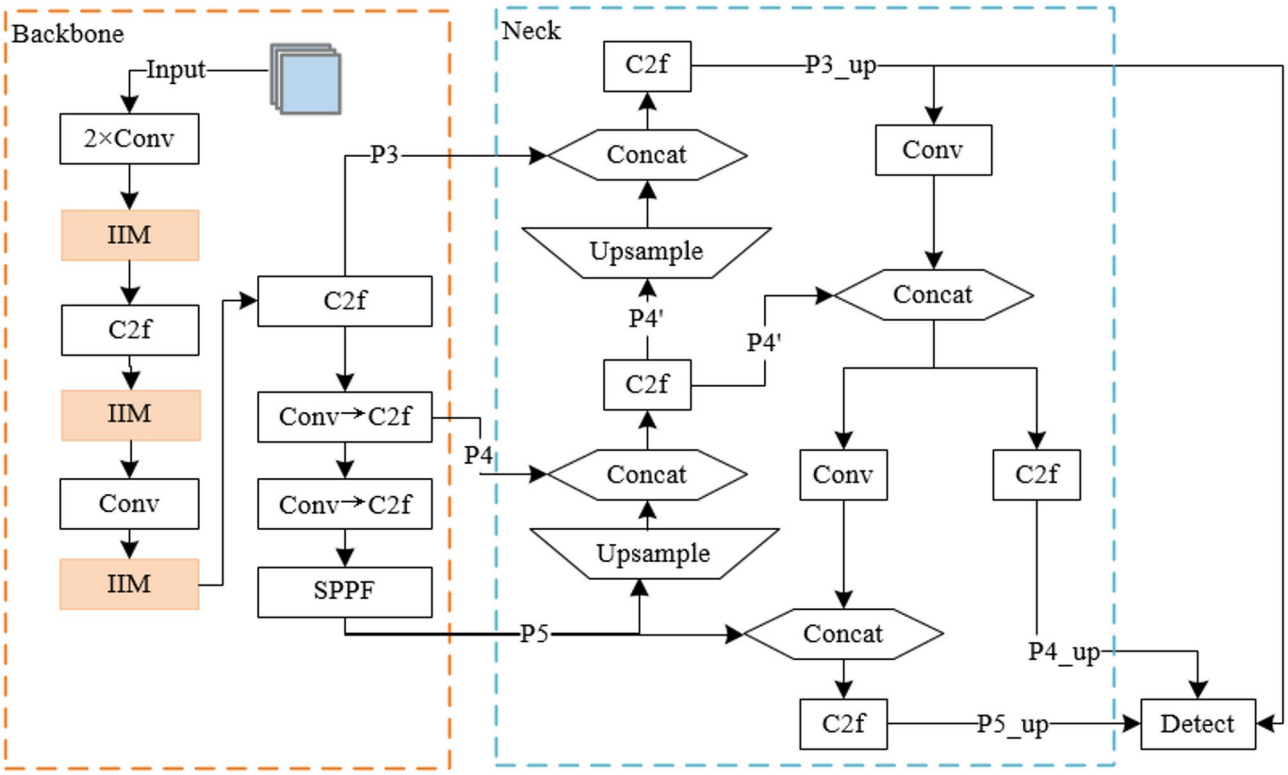


Fig. 4. Integration Locations of the IIM Module within the Network Architecture.

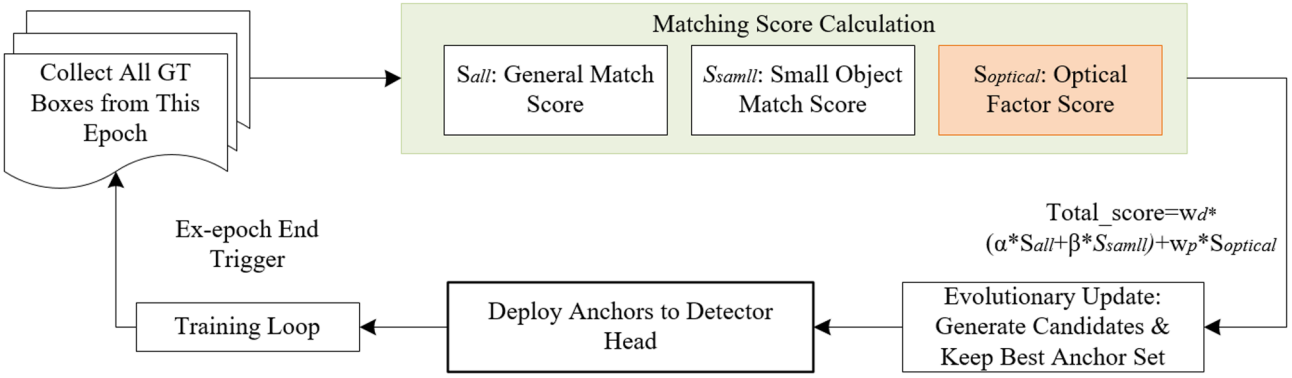


Fig. 5. AE-LFOG Anchor Optimization Workflow.

Comprehensive scoring function

The comprehensive scoring function *S(A)*is designed to simultaneously optimize the data matching degree and physical rationality:

$$\mathrm{S(A)}=\mathrm{w_d}\cdot\mathrm{S_{data}}+\mathrm{w_p}\cdot\mathrm{S_{optical}} \tag{3}$$

The data matching degree $\mathrm{S_{data}}$ ensures that optimized anchor boxes effectively cover ground truth targets in the training set. It is formulated as a weighted combination of global coverage and small-target sensitivity:

$$\mathrm{S_{data}}=\alpha\cdot\mathrm{S_{all}}+\beta\cdot\mathrm{S_{small}} \tag{4}$$

The global matching score $\mathrm{S_{all}}$ measures the overall coverage capability of the anchor boxes against all ground truth (GT) boxes. It is defined as:

$$\mathrm{S_{all}}=\frac{1}{\mathrm{N_{gt}}}\sum_{\mathrm{i}=1}^{\mathrm{N_{gt}}}\max_{\mathrm{j}\epsilon\{1,\ldots,\mathrm{N_a}\}}(\mathrm{IoU}(\mathrm{w_{gt,i}},\mathrm{h_{gt,i}},\mathrm{a_j},\mathrm{b_j})) \tag{5}$$

where $\mathrm{N_{gt}}$ is the total number of ground truth boxes, N_a is the total number of anchor boxes, $\mathrm{w_{gt,i}}$, and $\mathrm{h_{gt,i}}$ are the width and height (in pixels) of the *i*-th ground truth box, $\mathrm{a_j}$ and $\mathrm{b_j}$ are the width and height of the *j*-th anchor box, and IoU(·) denotes the Intersection over union function.

The small-target matching score $\mathrm{S_{small}}$,which evaluates coverage specifically for targets with area smaller than$(32/416)^2$, which is relative to input image size.

$$\mathrm{S_{small}}=\frac{1}{|\mathrm{G_{small}}|}\sum\nolimits_{\mathrm{i}\in\mathrm{G_{small}}}\max_{\mathrm{j}\in\{1,\ldots,\mathrm{N_a}\}}(\mathrm{IoU}(\mathrm{w_{gt,i}},\mathrm{h_{gt,i}},\mathrm{a_j},\mathrm{b_j})) \tag{6}$$

where $\mathrm{G_{small}}=\{\mathrm{i}|\mathrm{w_{gt,i}}\times\mathrm{h_{gt,i}}<(32/416)^2\}$ denotes the set of indices of all small target ground truth boxes.

In addition to data-driven metrics, the physical rationality Soptical incorporates modeling the imaging process of the camera system. This component evaluates whether an anchor box aligns with the theoretical projection of a real object through a thin-lens optical model. For an anchor box of a given pixel size, denoted as (a_{px}, b_{px}), we compute its expected physical projection (a_x,b_x) based on focal *f*, sensor physical size (in mm), sensor resolution, and object distance $\mathrm{D_{physics}}$. The score is defined as:

$$\mathrm{a_x}=\frac{\mathrm{W_{obj}}\cdot\mathrm{f}}{\mathrm{D_{physics}}},\quad \mathrm{b_x}=\frac{\mathrm{H_{obj}}\cdot\mathrm{f}}{\mathrm{D_{physics}}} \tag{7}$$

where $\mathrm{W_{obj}}$ and $\mathrm{H_{obj}}$ denote the real-world width and height of the target object The optical consistency score is then defined as:

$$\mathrm{S_{optical}}=1-\left|1-\frac{\mathrm{a_{px}}\mathrm{b_{px}}}{\mathrm{a_x}\mathrm{b_x}}\right| \tag{8}$$

A value closer to 1.0 indicates higher consistency with the physical imaging process, thereby enhancing the interpretability and robustness of the anchor design. In practical engineering deployments, the parameters $\mathrm{W_{obj}}$, $\mathrm{H_{obj}}$, *f* and especially the typical working distance $D_{physics}$ could be pre-configured based on site-specific conditions and loaded via external configuration files. This allows the AE-LFOG algorithm to be flexibly adapted to diverse monitoring scenarios without retraining, striking a balance between physical grounding and deployment versatility.

Learning strategy optimization

The design and weight configuration of the loss function play a decisive role in the convergence speed and final performance of deep learning models. In object detection tasks, the total loss is typically composed of multiple components including localization loss (L_{box}), classification loss (L_{cls}) and distribution focal loss (L_{df})[28].

$$\mathrm{L_{total}}=\lambda_{\mathrm{box}}\ \cdot \mathrm{L_{box}}+\lambda_{\mathrm{cls}}\ \cdot \mathrm{L_{cls}}+\lambda_{\mathrm{df}}\ \cdot \mathrm{L_{df}} \tag{9}$$

where λ_{box}, λ_{cls} and λ_{df} are hyperparameters that balance the contributions of each component. However, conventional approaches fix these weights throughout training, which overlooks the dynamic evolution of model learning requirements across different phases and under varying physical imaging conditions. However, such static weighting overlooks the dynamic evolution of model learning requirements across training phases and under varying physical imaging conditions.

To overcome this limitation, we propose a physics-guided adaptive loss weighting mechanism that couples directly with the AE-FLOG anchor optimizer. Instead of fixed or heuristic weights, it uses the imaging system's physical state as a real-time signal to dynamically adjust the balance between localization and classification losses. This creates a closed-loop framework that continuously adapts training to physical constraints. Figure 6 illustrates this mechanism.

As shown in Fig. 6, at the end of each training epoch, anchor boxes are first updated by the AE-LFOG optimizer. Subsequently, an image quality score $\mathrm{S_{optical}}$ is computed to reflect the fidelity of the input. Lower $\mathrm{S_{optical}}$ indicate degraded imaging conditions, including blur, distortion, or low illumination, which harm localization accuracy. Thus the bounding box regression loss weight λ_{box} is increased to prioritize bounding box regression. Conversely, higher $\mathrm{S_{optical}}$ values reflect favorable conditions, allowing the classification loss weight λ_{cls} to dominate. The dynamic weighting is governed by Eqs. (10) and (11).

$$\lambda_{\mathrm{box}}^{(\mathrm{t}+1)}=\lambda_{\mathrm{box}}^{(0)}\cdot\ [1+\alpha\ \cdot(1-\mathrm{S_{optical}}^{(\mathrm{t})})] \tag{10}$$

$$\lambda_{\mathrm{cls}}^{(\mathrm{t}+1)}=\lambda_{\mathrm{cls}}^{(0)}\cdot\ [1-\beta\ \cdot(1-\mathrm{S_{optical}}^{(\mathrm{t})})] \tag{11}$$

where t denotes the current epoch, $\lambda_{\mathrm{box}}^{(0)}$ and $\lambda_{\mathrm{cls}}^{(0)}$ are the base loss weights. The α and β are scaling factors that control the sensitivity. The entire process is embedded via a callback architecture, ensuring seamless integration into the training pipeline without interrupting gradient flow. By closing the loop between physical sensing and learning dynamics, this approach enhances robustness across heterogeneous environments.

Experiments and analysis

The proposed illumination-robust detection method with dual optimization strategy (denoted as Ours) was comprehensively evaluated on both specialized construction site datasets with challenging lighting conditions and multi-scale objects, as well as on public benchmarks.

Experimental setup

All experiments were conducted under the unified configuration detailed in Table 1. Performance was evaluated using standard object detection metrics as mAP@0.5, which is mean Average Precision(AP) at IoU threshold 0.5 and mAP@0.5:0.95 that is mean AP averaged over IoU thresholds from 0.5 to 0.95 with a step size of 0.05. In this paper, N = 10, and AP represents the average AP across all classes at the i-th IoU threshold. Precision ($\mathrm{P}=\frac{\mathrm{TP}}{\mathrm{TP+FP}}$), Recall ($\mathrm{R}=\frac{\mathrm{TP}}{\mathrm{TP+FN}}$) and F1-score ($\mathrm{F1}=2\times\frac{\mathrm{P\times R}}{\mathrm{P+R}}$) are common evaluation indicators of classification task[8], where TP denotes the number of correctly detected objects, FP denotes the number of erroneous detection, FN means the number of missed detection.

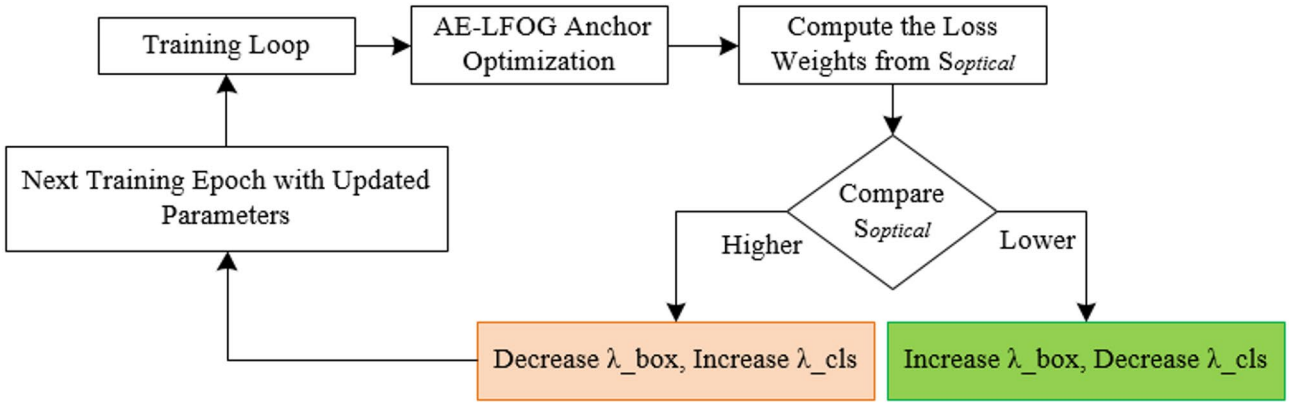


Fig. 6. Framework Diagram of the Adjustment Mechanism.

Parameter	Configuration	Parameter	Configuration
Hardware Platform	Linux Server	Initial Learning Rate	0.01
GPU	NVIDIA RTX 3090 (24GB VRAM)	Learning Rate Scheduler	Cosine Annealing
Software Framework	Python 3.10.18	Momentum	0.937
Deep Learning Framework	torch 2.3.1	Weight Decay	5×10^{-4}
CUDA Version	12.8	Data Augmentation	Mosaic, MixUp, Horizontal Flip
cuDNN Version	8.9.5	Key Hyperparameters	$\beta = 0.1$, $\alpha = 0.5$
Batch Size	32	Anchor Boxes	9 anchors (3 per scale)

Table 1. Experimental environment and training configuration.

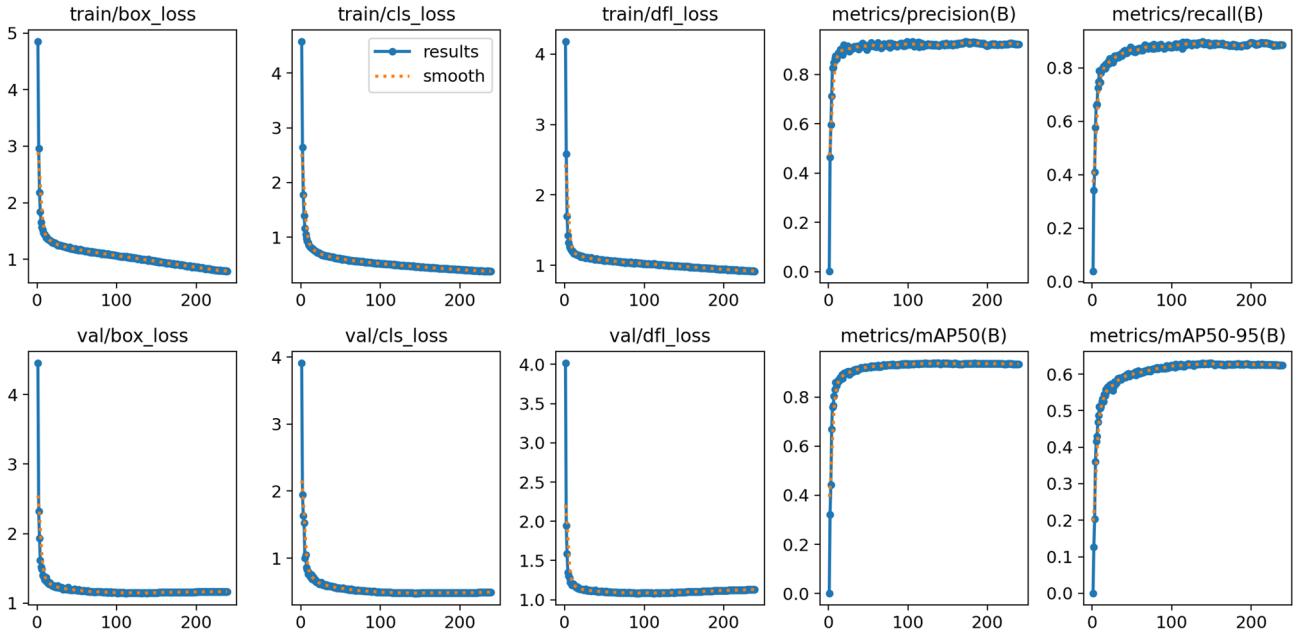


Fig. 7. Variation trends of loss functions and evaluation metrics during model training.

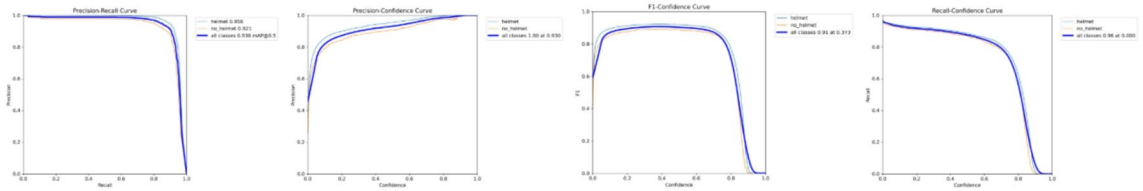


Fig. 8. Multidimensional Performance Evaluation Curves.

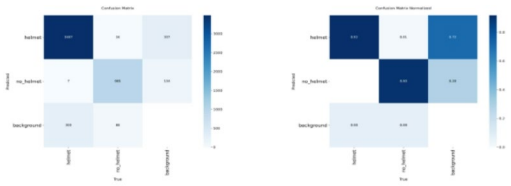
Experimental results

The proposed method was evaluated on standard benchmarks. Training convergence as Fig. 7 and validation performance as Fig. 8 demonstrate stable optimization, achieving mAP@0.5 of 0.948 and peak F1-score of 0.91. The mAP@0.5:0.95 value reflects consistent performance across IoU thresholds. Figure 9 further shows high precision for both helmet and no-helmet classes. Figure 9 shows the confusion matrices (raw and normalized), revealing discrimination between helmet and other classes.

Figure 10 demonstrates the helmet detection performance of the proposed model across diverse practical scenarios. In dense and complex scenes, our method exhibits a lower false positive rate; the baseline model misclassifies circular objects or ordinary hats as safety helmets.

Ablation studies

Ablation studies conducted on the public datasets systematically evaluate the contributions of each component in the proposed framework, with detailed results summarized in Table 2.



(a) (b)

Fig. 9. Confusion matrix of the model on the tunnel safety helmet detection dataset. (**a**) Raw counts; (**b**) Normalized ratios.



(a) (b)

(c) (d)

Fig. 10. Visualisation results. Comparing YOLOv8m (left) and our model (right).

Model	YOLOv8m (Baseline)	YOLOv8m + AE-LFOG	YOLOv8m + IIM	Ours(YOLOv8m + AE-LFOG + IIM)
mAP@0.5	93.41%	94.52%	93.53%	**94.83%**
mAP@0.5:0.95	62.13%	63.35%	62.73%	**64.18%**
Precision	92.04%	93.03%	91.71%	**93.22%**
Recall	88.63%	88.68%	89.08%	**89.54%**
val/box_loss	1.16	1.16	1.23	**1.15**
val/cls_loss	0.5	0.48	0.52	**0.47**
FPS	90.17	81.66	76.66	**120.99**
GFLOPs	39.53	39.53	43.1	**6.90**
Avg Inference Latency	11.09ms	12.25ms	13.04ms	**8.26ms**
Parameters	**25.86**	**25.86**	26.15	26.15

Table 2. Ablation experiment. The best performance parameters are in bold.

The ablation study demonstrates that both proposed modules contribute to performance gains. The AE-LFOG module improves mAP metrics, indicating enhanced feature decoupling and localization accuracy. The IIM module boosts recall, validating its efficacy in handling complex illumination. The full model achieves the best results across all metrics, with + 2.05% in mAP@0.5:0.95 over the baseline, confirming the complementary benefits of integrating both modules.To enhance the reliability of the model generalization ability evaluation and reduce the bias introduced by data partitioning, we adopted k-fold cross-validation to assess the model, with the results presented in Table 3.

Performance comparison

To comprehensively evaluate the performance of the proposed method, we compared it with four recently proposed representative approaches.

Among models with full performance metrics as Table 4, our method achieves the highest mAP@0.5 of 94.83% and mAP@0.5:0.95 of 64.18%, surpassing the YOLOv8m baseline by 1.42 and 2.05% points, respectively,

Algorithm	AE-LFOG-YOLO				
K	1	2	3	4	5
mAP@0.5	94.75%	94.43%	94.50%	93.38%	94.19%

Table 3. Results of k-fold cross-validation.

Metric	mAP@0.5	mAP@0.5:0.95	Precision (*P*)	Recall (*R*)	FPS	Para
Ours	94.83%	64.18%	93.22%	89.54%	120.99	26.15
YOLOv8m(Baseline)	93.41%	62.13%	92.03%	88.63%	90.17	25.86
IYA [22]	93.90%	63%	89.40%	87.50%	-	-
URD-YOLOv8[23]	91.51%	57.90%	92.47%	85.46%	109	-
YOLOv8s-SNC[24]	92.60%	56.00%	92.40%	86.60%	44	14.5
BLP-YOLOv10[25]	98.10%	-	94.50%	99.00%	-	3.28
YOLO-LHD[26]	94.30%	58.60%	92.40%	88.80%	96.15	0.86
Transformer + YOLOv10[27]	83.43%	80.23%	-	-	37.8	-

Table 4. Comparison of different algorithms.

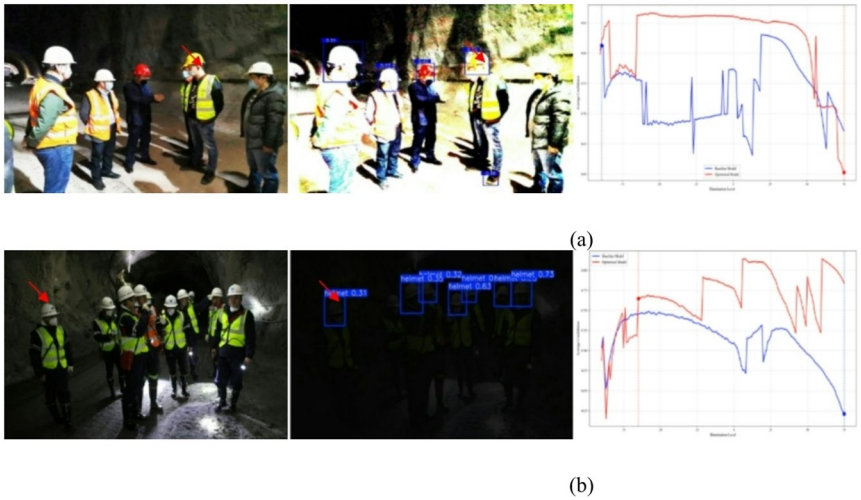


Fig. 11. Experimental results for robustness evaluation: (**a**) Class A scenario featuring multi-scale targets under varying illumination; (**b**) Class B scenario featuring high-density crowded scenes under adverse illumination.

at 120.99 FPS and 26.15 million parameters. It outperforms IYA and URD-YOLOv8 in precision and recall, and runs approximately three times faster than Transformer with YOLOv10. Although BLP-YOLOv10 reports a higher mAP@0.5 of 98.10%, it omits mAP@0.5:0.95 and inference speed, precluding a comprehensive comparison.

Robustness experiments

This study presents a comparative evaluation between the baseline YOLOv8m and the proposed optimized model. Illumination levels were systematically varied from −90 to +75 via gamma transformation as $\gamma = 1 + \frac{L}{100}$, where negative values brighten and positive values darken the image, simulating extreme lighting conditions. Evaluation metrics include average confidence, effective operating range (EOR) defined as the illumination range with confidence > 0.6), average detection count within EOR, and small object detection count (< 400 pixels²) at L = −50, collectively enabling a comprehensive analysis of illumination robustness.

To evaluate robustness, two test scenarios were defined as Class A features multi-scale targets, and Class B features high-density crowded scenes under adverse illumination. For each scenario, we generated 166 illumination variants via gamma transformation to simulate extreme conditions.

As shown in Fig. 11, our model maintains detection confidence across a wide range of illumination levels, with performance degrading only under extreme situation. As detailed in Fig. 11 (a) and (b), the model begins to miss-detection (marked by red arrow) when blur intensity exceeds level 71 or illumination drops below level 68.

Metric	Baseline Model	Optimized Model	Improvement
Average Confidence	0.724	0.786	8.6%
Lower Bound of EOR	-20	-35	15%
Upper Bound of EOR	50	60	10%
Total EOR Span	70	95	35.7%
Average Count in EOR	8.7	10.3	18.4%
Small Objects Count	6.2	7.5	21%

Table 5. Comprehensive performance comparison between the baseline model and the optimized model.

The accompanying curves quantify this performance drop-off at these failure points. A comparison with baseline models is provided in Table 5.

As shown in Table 5, the optimized model exhibits improved illumination robustness compared to the baseline, with an average detection confidence of 0.786 versus 0.724 across brightness levels. The effective operating range extends from 70 to 95, representing a 35.7% expansion, and its bounds shift from −20 and +50 to −35 and +60. Within this range, it detects 18.4% more objects on average and 21% more small objects, suggesting enhanced consistency under varying lighting conditions.

Conclusion

This paper presents AE-LFOG-YOLO, a robust framework for safety helmet detection in complex illumination environments like tunnel construction. It integrates physical optics with deep learning: IIM suppresses lighting artifacts via adaptive feature recalibration, while AE-LFOG dynamically optimizes anchors using imaging priors. Evaluated on a real-world tunnel dataset, it achieves mAP@0.5 of 94.83%. Deployed on Chongqing Rail Transit Line 6 for over six months, the system demonstrates practical efficacy. Future work may explore model compression for edge deployment and semi-supervised adaptation to reduce annotation dependency.

Data availability

The data used in this paper is public that named as Safety Helmet Detection and has been deposited at https://tianchi.aliyun.com/dataset/94696. And dataset named as VOC2028 has been deposited at https://aistudio.baidu.com/datasetdetail/43757. The datasets used and analysed during the current study available from the correspon ding author on reason able request.

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Author contributions
Suimei Liu designed the study and wrote the main manuscript text and prepared all figures. Jun Wang supervised the research. All authors reviewed the manuscript.

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Declarations

Competing interests
The authors declare no competing interests.

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